GLOBAL WORKFORCE REDUCTIONS 2020–2023

An exploratory SQL analysis of company, industry and geographic trends

Prepared by: Anchal Chauhan | **Submission:** 30 November 2026

1. Introduction

Workforce reductions are an important indicator of organisational restructuring, economic pressure and changes in business priorities. This study examines a publicly shared dataset of company layoff announcements to identify where the largest reported reductions occurred and how their distribution changed between March 2020 and March 2023. The objective is descriptive: to uncover patterns in the available records without assuming that they represent every layoff worldwide.

2. Research questions

Which years saw the largest reported reductions? Which industries, countries and companies contributed most to recorded layoffs? What does the monthly pattern reveal, and what limitations should guide interpretation?

3. Dataset and analytical approach

The supplied CSV contains **2,361 rows** and nine columns covering company, location, industry, number and percentage laid off, announcement date, funding stage, country and funds raised. The recorded dates span **11 March 2020 to 6 March 2023**. A duplicate-record check identified **5** exact duplicates; calculations in this report remove those duplicates. Missing layoff counts are excluded from numerical totals rather than treated as zero. Industry labels beginning with “Crypto” were grouped together, and country punctuation was standardised.

The accompanying SQL script demonstrates staging tables, ROW_NUMBER() duplicate checks, TRIM(), date conversion, joins for missing industry values, GROUP BY summaries, CTEs and window-function analysis. For reproducibility, the numerical results and figures in this report were independently calculated from the supplied CSV using the explicitly stated cleaning rules; the SQL script was reviewed but not executed against a MySQL database.

4. Executive findings

2,361	383,659	2020–2023
Source records	Reported layoffs	Observed years

The largest annual total occurs in **2022 (160,661)**. **Consumer** leads industry totals (44,782), while the **United States** accounts for 66.9% of the reported total. **Amazon** has the highest cumulative company total (18,150). These are counts in the dataset, not population-wide layoff rates.

5. Findings: timing and scale

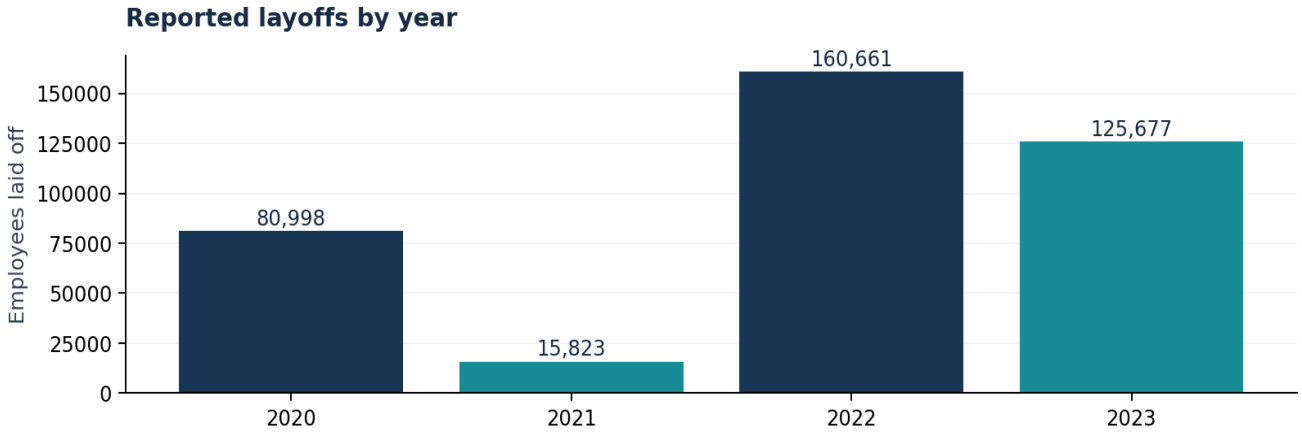


Figure 1. Annual sums from non-missing layoff counts. 2023 is partial (through 6 March), so comparisons with full years require caution.

The data show a sharp rise in recorded layoffs from 2021 to 2022. A substantial number were also recorded in early 2023. Because the dataset ends in March 2023, the 2023 figure should not be interpreted as a full-year result.

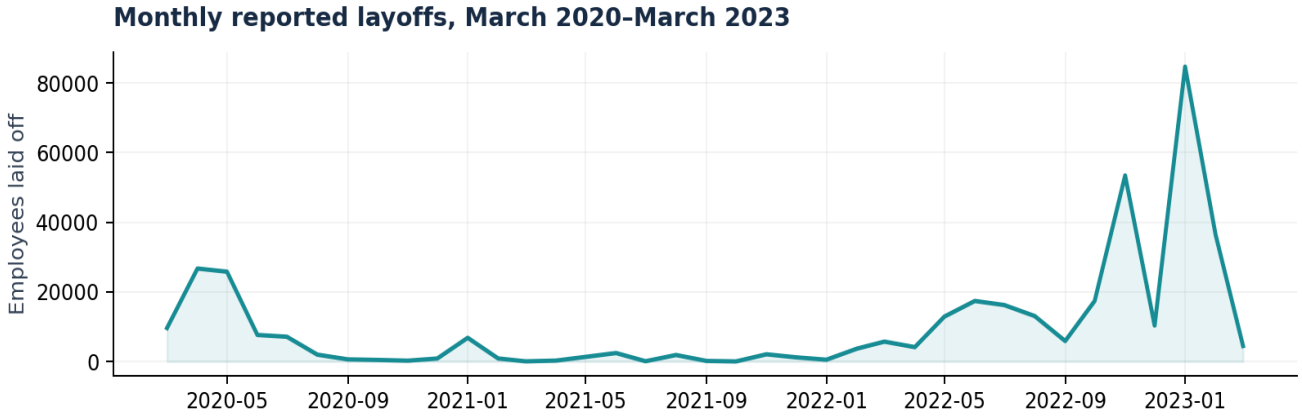


Figure 2. Monthly totals are aggregated by announcement date; changes can reflect reporting coverage and the size of individual announcements.

Monthly totals are uneven, with noticeable spikes rather than a smooth upward or downward progression. Large individual company announcements can influence these peaks, so month-to-month changes should not be read as evidence of a broader labour-market trend without external benchmarks.

6. Findings: industry and geography

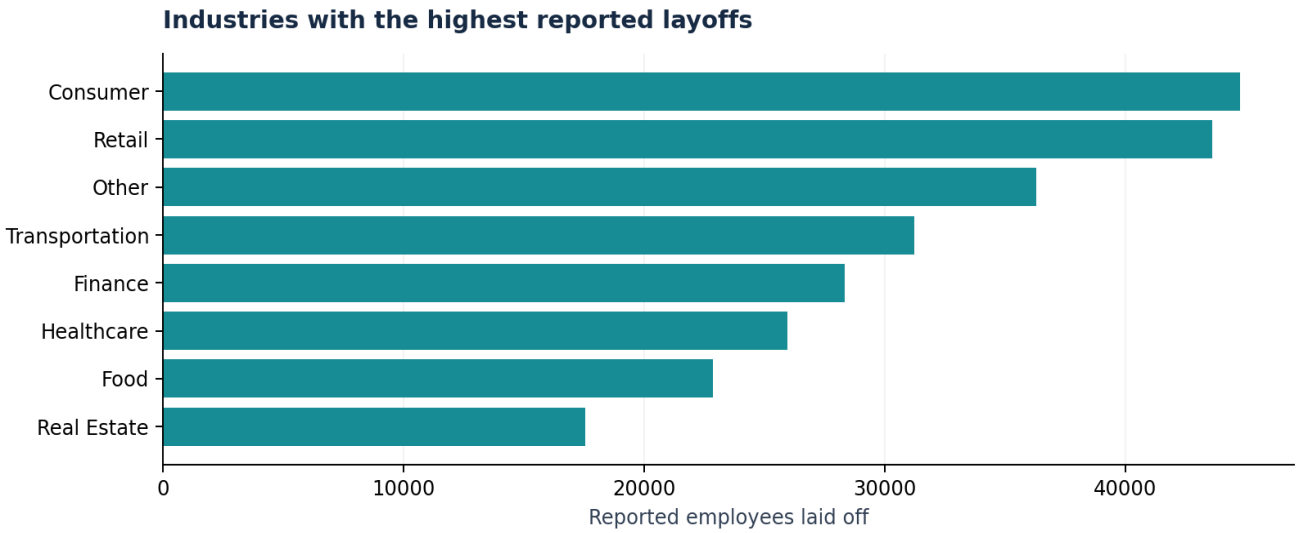


Figure 3. The eight industries with the largest reported employee counts.

Consumer (44,782) and Retail (43,613) rank highest in this dataset. These comparisons reflect the absolute number of employees reported as laid off, not the share of each industry workforce.

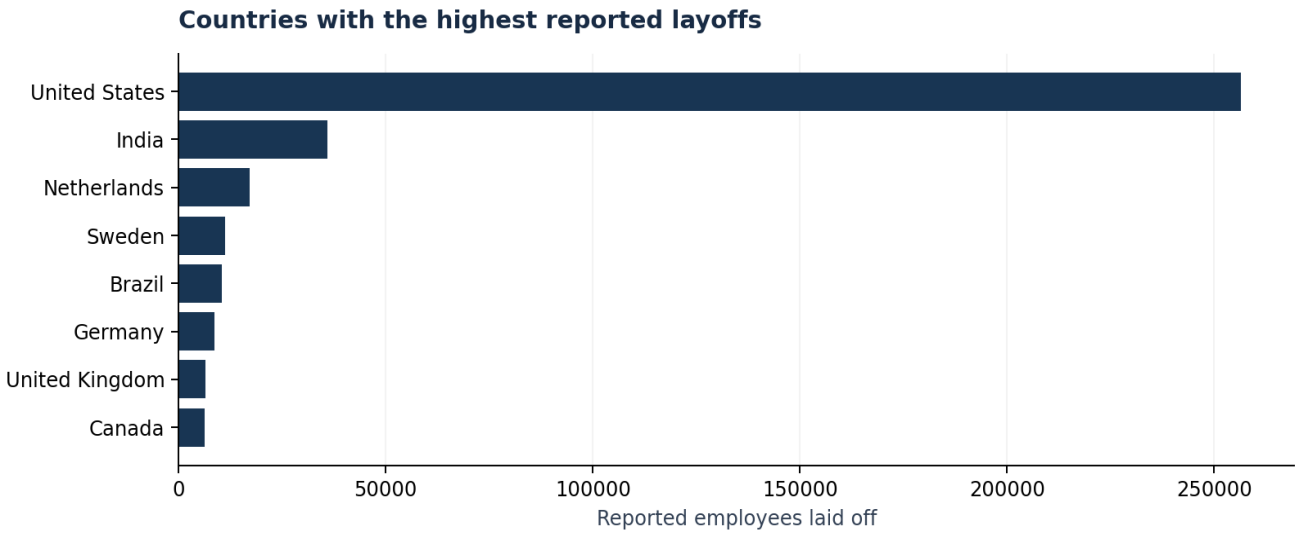


Figure 4. Country totals reflect where recorded announcements are attributed, not comparable national unemployment rates.

The United States contributes 256,559 reported layoffs, considerably more than the next-ranked countries. This concentration may reflect both actual large announcements and differences in dataset coverage.

7. Findings: company concentration

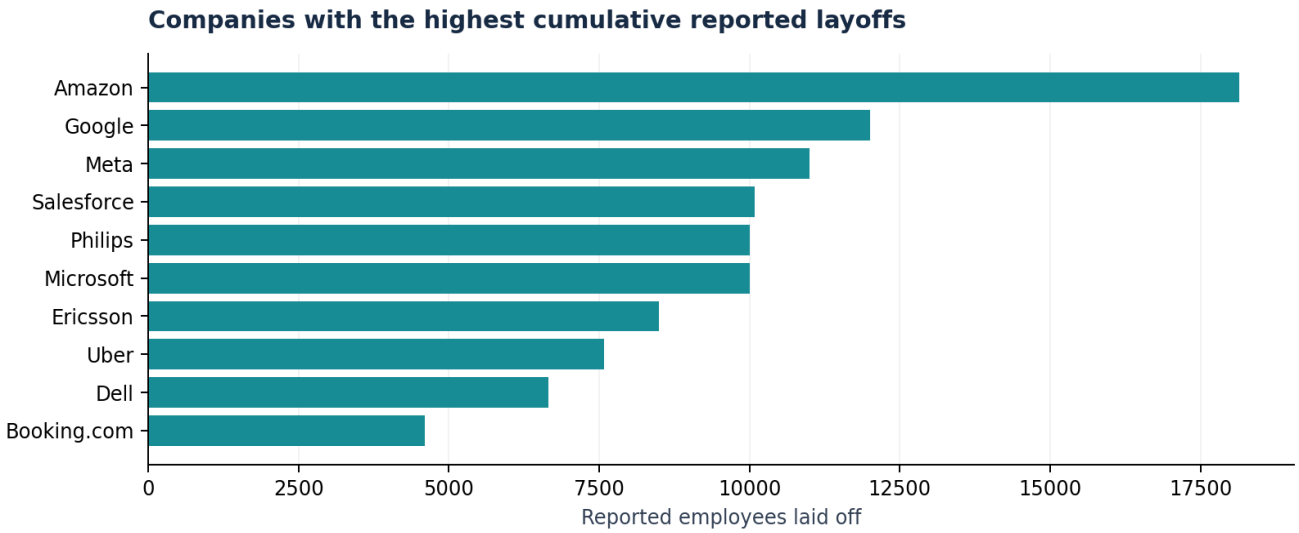


Figure 5. Top ten companies ranked by the sum of their recorded layoff counts across announcements.

Company totals are aggregated across available records, so they may reflect multiple announcements by the same employer. They should not be interpreted as distinct employees without checking the underlying events.

8. Data quality and interpretation

Of the 2,361 source rows, 740 have no reported numerical layoff count, and 785 have no percentage. Missing values restrict the completeness of the sums. The dataset is not a random sample of employers and does not include workforce denominators, so it cannot establish layoff risk or causal explanations. The supplied chart titled “Yearly Top 5 Companies by Layoffs” uses a year axis in a way that could be misleading; the figures here use actual dates and separate company rankings.

9. Recommendations and next steps

For a business-facing dashboard, separate absolute layoff counts from layoff percentages, provide year and country filters, and mark partial periods prominently. To strengthen the analysis, join the dataset with industry employment totals, company headcount and external economic indicators; then assess whether apparent differences persist after accounting for size and reporting coverage.

10. Conclusion

The analysis identifies a pronounced concentration of recorded workforce reductions in 2022 and early 2023, particularly among large employers and certain consumer-facing industries. SQL provides a clear framework for cleaning, grouping and ranking these records. The strongest conclusion is descriptive: the dataset reveals where reported layoffs were concentrated, while incomplete fields and uneven coverage limit broader generalisation.

Data source and reproducibility

The dataset used in this analysis was obtained from Kaggle and is based on publicly available layoff records compiled by Layoffs.fyi. It covers workforce reductions across companies, industries and countries between 2020 and 2023.
Original source: https://layoffs.fyi/
Dataset: https://www.kaggle.com/datasets/swaptr/layoffs-2022